1. Moddiy nuqtaning harakat tenglamalari $x=5+3t$ va $y=3+\sqrt{3}t$ ko‘rinishida berilgan. Moddiy nuqataning harakat yo‘nalishi *Ox* o‘qi bilan qanday burchak tashkil qiladi?

Berilgan: $x_0=5$, $y_0=3$;

Topish kerak: α-?

Masalaning yechilishi:

t = 0 vaqt momentida nuqtaning koordinatalari $x_0=5$, $y_0=3$. Biror vaqtdan so‘ng, masalan t = 1 s da moddiy nuqta koordinatalari $x_1=8$ va $y_1=3+\sqrt{3}$ bo‘lgan vaziyatda bo‘ladi. Ma’lumki, nuqtaning ikki vaziyatini tutashtiruvchi to‘g‘ri chiziqni *Ox* o‘qi bilan tashkil etgan burchagi

$$tg\,\alpha=\frac{y_1-y_0}{x_1-x_0}$$

ifoda bilan aniqlanadi. Moddiy nuqta kkordinatalarining qiymati asosida hisoblash ishkarini bajaramiz:

$$tg\,\alpha=\frac{3+\sqrt{3}-3}{8-5}=\frac{1}{\sqrt{3}};$$

Masalaning javobi:$\alpha=30^0$.

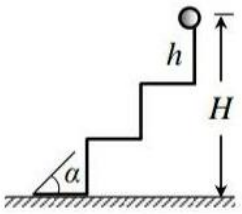
2. Koptok zinapoyaning balandligi 0,9 m bo‘lgan 3 ta po‘g‘onasidan dumalab tushdi. Har bir pog‘onaning balandligi (h) uning eniga teng. Zinapoyaning gorizontga qiyaligi 45^0. Koptokning yo‘li va ko‘chishini toping.

Berilgan: $H=0{,}9$ m, $\alpha=45^0$;

Topish kerak: s - ? Δl - ?

Masalaning yechilishi:

Masala chizmasidan zinaning balandligi $h = H/3$ ekanligini ko‘rish mumkin. Agar koptok sakramasdan dumalasa, uning bosib o‘tgan yo‘li



pog‘ona balandligi va eni yig‘indisining uchlanganiga teng:

$$s = 3\left(\frac{H}{3} + \frac{H}{2}\right) = 2H \qquad (1)$$

Koptokning boshlang‘ich va oxirgi vaziyatlarini tutashtiruvchi to‘g‘ri chiziq Δl (ko‘chish) deb olinsa, 1.1-rasmdan

$$\sin \propto = \frac{H}{\Delta l} \qquad (2)$$

bundan

$$\Delta l = \frac{H}{\sin \propto} \qquad (3)$$

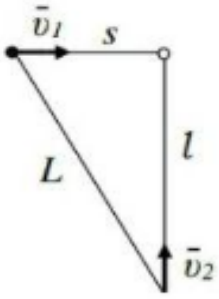
Hisoblash natijalari: s = 1,8 *m*, Δl = 1,27 *m.*

3. Ovchi o‘zidan 30 m uzoqdagi qushga quroldan o‘qotmoqda. O‘q qushning harakat traektoriyasiga perpendikulyar yo‘nalishda harakatlanadi. Qushning tezligi 15m/s, sochmaning tezligi 375m/s. Qush o‘q otilgandan sochma unga tekkunga qadar qancha yo‘lni bosib o‘tadi?

Berilgan: L = 30 *m*, v_1 = 15 *m/s*, v_2 = 375 *m/s*;

Topish kerak: s - ?

Masalaning yechilishi:



Qush va o‘qning harakati uchun mos ravishda quyidagi tenglamalarni yozamiz:

$s = v_1 t$ (1)

$l = v_2 t$ (2)

Pifagor teoremasidan o‘qning harakati masofasi:

$$l = \sqrt{L^2 - s^2}$$

Oxirgi tenglikni e’tiborga olib, (1) va (2) ifodalarning nisbatini olamiz:

$$\frac{s}{v_1} = \frac{\sqrt{L^2 - s^2}}{v_2}$$

bundan

$$s = \frac{v_1 L}{\sqrt{v_1^2 + v_2^2}} \qquad (3)$$

Hisoblash natijasi: s = 1,19 m.

@olimpdep
Fan olimpiadalari bo‘yicha
iqtidorli o‘quvchilar bilan ishlash
DEPARTAMENTI